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## WS2025/26 Modeling Agents: Final Projects

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The main goal of the projects as the practical part of the course is to give you an opportunity to explore the following broad questions in hands-on projects that have been addressed in discussion throughout the course (also see slides from session 1 for the general framing):

- Using ideas of computational cognitive science, how can we reverse-engineer complex systems (like human minds complex AI systems) to better understand their behavior and mechanisms?
- Specifically, which factors and mechanisms support complex behaviours like decision making, planning, theory of mind, language, social reasoning, learning and more?

Given that the focus of the course was (mostly) on Bayesian models of human cognition, the idea of the projects is to use those Bayesian models to better understand other kinds of complex systems that seem to exhibit non-trivial complex behaviors — language models (LMs). The reason for focusing on LMs instead of humans is that we do not have the logistics for collecting data from humans, while (some) experience in working with LMs is assumed for most participants of the course and their predictions are more easily accessible.

In the following, you will first find general instructions that apply to all projects, then a proposal for the project research question / outline and then a list of papers based on which the research question can be investigated. If you are working on your own project idea, please still read through the following guidelines to make sure you don't miss any general information. Specific details on grading and formal requirements (length of write up, its structure etc) can be additionally found on Moodle, under the section Projects > "Project requirements & grading criteria". In particular, requirements for a proposal for your own projects can be found there and should be followed as described.

Importantly, the research questions are outlined on a high level; filling in the concrete experimental details are part of your task for working on the project. Additionally, the projects do not have a "ground truth solution", but are rather intended as an opportunity for you to build your independent research skills, critical thinking and practice clearly reporting your ideas and experiments to an uninformed reader. The outline and research question below are a starting point; you can follow it more closely, but you are also free to choose alternative ideas and research questions, as long as you explain and motivate them clearly in your write-up.

### GENERAL INSTRUCTIONS

- (If applicable) deadline for submission of own project proposals: **February 28th 2026**
- Deadline for project submission: **April 15th 2026**
- Your projects should be completed in groups of up to 5 students. Individual submissions will not be accepted. Any changes to the groups relative to the sign-up on Moodle, or any issues within the group should be communicated to me as soon as possible.
- Your task is to work on your selected project, write up the results in a conference-style paper, submit the paper, and a clean repository with all the code and results. A single PDF / zip with the project should be submitted through Moodle. Please see the guidelines on Moodle for details.
- For materials on working with LMs, if needed, please see the webbook from the [Understanding Language Models](#) course

- If you need help, please try to ask your group mates or other course participants. If you cannot reasonably work with open-source LMs and need API keys to access closed-source LMs, please contact me.
- I highly recommend that you do not hesitate to reach out to me via email to discuss any open questions, your ideas, ... as soon as possible (but please do try to formulate maximally concrete questions or provide examples of your difficulties, and account for some response time on my side, please).

## PROJECT STRUCTURE AND RESEARCH QUESTION

- Please select a paper (e.g., see the list of suggested papers below) that proposes a Bayesian model explaining human behavior on a certain task and provides results of human performance on that task. The high-level idea is to 1) compare the LM's performance to humans results reported in the paper, and 2) use the computational model that is proposed in the paper and is shown to explain human data well in order to investigate the research question: is the LM behavior also explained well by the same computational model (i.e., how human-like is the LM's performance)?
- (1) Comparison to human performance:
  - As the first step, please replicate the experiment(s) reported in the paper with LMs. This may require comping up with the experimental materials based on the descriptions / examples from the paper, or e.g. casting them into text modality in order to test text-based LMs. If materials are made available by the paper authors, you may reuse them.
  - Please evaluate a selected LM (ideally, open-source) on the task and compare its performance to human behavior. Please see the webbook referenced above to see best practices and approaches to evaluations of LMs.
- (2) Using the Bayesian computational model to assess the LM's mechanism underlying the task performance:
  - The Bayesian model will likely have several components (e.g., a prior, a likelihood component). The general idea is that performing a target task according to the mechanism specified by the Bayesian model requires accurately representing these components and putting them together according to the formula specified by the model. Therefore, to investigate if the LM performs the task in accordance with the Bayesian model, please investigate if the language model "matches" (i.e., accurately represents) the components of the Bayesian model.
  - That is, construct an experiment to evaluate whether the model e.g. predict the same prior probabilities or likelihoods as assumed / specified in the paper. This can take the form of e.g. prompting the LM to predict probabilities of different observations that were used in the experiment in step (1) and which the Bayesian mechanism is supposed to explain. Please evaluate how closely / accurately the model predicts the relevant information.
- To help you think through how to answer the research question, here are four possible (stylized) outcomes of your evaluations:
  - O1: LM matches (or exceeds) human performance on the task and accurately matches the information of the Bayesian model components -> this pattern suggests that the Bayesian model also explains LM behaviour well.
  - O2: LM matches (or exceeds) human performance on the task but does *not* match the information of the Bayesian model components well -> this pattern suggests that some other mechanisms than proposed by the Bayesian model are at play in the LM.
  - O3: LM does not match human performance on the task but does match the information of the Bayesian model components well -> this pattern suggests that the model fails to implement the normative inference mechanism on top of these components, and that may lead to the failure on the task.
  - O4: LM does not match human performance on the task and does not match components -> this pattern suggests that failure on task may be due to failure to represent the required

subcomponents like prior and likelihood, although it does not rule out that the failure on the task is due to something else.

To help you imagine what this outline and research question may look like applied to a concrete task and computational model, please see the following example papers. These are intended as inspiration, please do *not* replicate the papers one to one. But you are welcome to take close inspiration from the procedure and e.g. apply them to a different dataset / task. The papers intended for your actual project are listed after that.

Here are the papers to illustrate this kind of approach:

- Murthy et al. (2025): Using Cognitive Models to Reveal Value Trade-offs in Language Models
- Zhu & Griffiths (2024): Eliciting the Priors of Large Language Models using Iterated In-Context Learning
- Tsvilodub et al. (2025): Non-literal Understanding of Number Words by Language Models

#### **SUGGESTED PAPERS FOR PROJECTS ACCORDING TO THE SCHEME ABOVE**

- Velez-Ginorio et al. (2017): Interpreting actions by attributing compositional desires.
- Goodman & Stuhlmüller (2013): Knowledge and Implicature: Modeling Language Understanding as Social Cognition
- Lieder et al. (2018): Empirical evidence for resource-rational anchoring and adjustment
- Jara-Ettinger et al. (2016): The Naïve Utility Calculus unifies spatial and statistical routes to preference.